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The moderating role of academic integrity in the relationship between artificial intelligence and critical thinking among graduate students in Jordan

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The differential effects of artificial intelligence (AI) on critical thinking in postgraduate education, particularly in the context of academic integrity, remain underexplored. This cross-sectional study addressed this gap by surveying a convenience sample of 555 postgraduate students from a public university in Amman, Jordan. Results indicated that AI usage positively supports lower-order cognitive skills (remembering, understanding, applying), whereas its impact on higher-order skills (analyzing, evaluating, creating) was weaker and contingent on students' academic integrity. Structural equation modeling demonstrated that academic integrity moderates the relationship between AI usage and critical thinking: students with high integrity effectively leveraged AI to enhance higher-order skills through deep engagement, critical evaluation, and idea generation, while students with lower integrity showed minimal gains in these skills, often relying superficially on AI outputs. For lower-order skills, all students benefited, though those with higher integrity achieved more meaningful improvements. These findings highlight that responsible and ethically guided AI use, reinforced by strong academic integrity, is essential for fostering both foundational and advanced critical thinking in postgraduate education. The study offers theoretical and practical insights for educators, policymakers, and researchers seeking to integrate AI effectively and ethically in higher education.

KEYWORDS

academic integrity, artificial intelligence (AI), critical thinking, Jordan, postgraduate students

Background

Amid rapid technological transformations, artificial intelligence (AI) has emerged as a key branch of computer science, capable of simulating human cognitive functions such as learning, decision-making, and problem-solving. Generative AI, particularly large language models like ChatGPT developed by OpenAI, represents a transformative advancement by generating novel content—including text, images, and code—beyond traditional data analysis. These tools have become widely accessible, notably within higher education, contributing to digital transformations in teaching, learning, and research. Other platforms, such as Google Gemini, Grammarly AI, Jasper, and Perplexity, similarly offer advanced generative and creative capabilities, supporting

academic writing, knowledge acquisition, and analytical synthesis (Gray et al., 2025; Gammoh, 2024; Altikriti and Nemrawi, 2025).

Artificial intelligence (AI) has reshaped critical aspects of contemporary life, notably in higher education, where it drives innovation in administrative, research, and pedagogical practices. The emergence of generative AI tools, such as ChatGPT and other OpenAI applications, has enhanced learning experiences, expanded opportunities for specialized education, and supported teaching and administrative processes. Universities increasingly utilize intelligent teaching applications and virtual assistants for predictive analytics, automated assessment, and personalized learning, while also employing AI in academic research through data analysis, content generation, literature reviews, and scholarly writing. Studies across universities in Jordan, Saudi Arabia, and the UAE indicate that around 88% of students use AI-based virtual assistants for assignments, examinations, language learning, and programming (Aldossary et al., 2024; Almulla and Ibrahim Ali, 2024; Alnsour et al., 2025; Al Kaabi, 2025; Al Mashagbeh et al., 2025; Vieriu and Petrea, 2025). Furthermore, AI is leveraged to promote self-directed learning among postgraduate students, support curriculum development, and enhance adaptive and digital learning platforms (Almulla and Ibrahim Ali, 2024; Alnsour et al., 2025).

AI has demonstrated a notable interactive presence among postgraduate students through the use of its tools in developing research ideas, performing spelling and grammar checks, paraphrasing, ensuring textual coherence, and identifying prior studies and relevant literature related to their research topics. This has contributed to the expanded adoption of these technologies within higher education institutions, alongside their role in enhancing academic writing and data analysis processes. Such developments have been reflected in increased research productivity and improved academic performance efficiency (Anani et al., 2025; Vincent et al., 2025).

Despite the multiple positive contributions of AI in supporting the educational process—particularly in personalized learning, improving academic outcomes, and enhancing student engagement—its use simultaneously raises a range of challenges related to output reliability and accuracy, as well as emerging ethical concerns in the context of higher education (Essien et al., 2024; Vieriu and Petrea, 2025; Gammoh, 2024). Excessive reliance on AI tools has been associated with increased academic misconduct, deterioration of intellectual originality, and a decline in the quality of university assignments, which may ultimately threaten graduates' workforce readiness and undermine employers' confidence in university outputs (Al Kaabi, 2025). Al Kaabi (2025) examined the impact of university students' reliance on artificial intelligence tools in completing academic tasks on critical thinking skills, academic integrity, and employability readiness in higher education institutions in the United Arab Emirates. Using a social survey methodology and a stratified random sample of undergraduate students from higher technical colleges, the findings indicated that the majority of students relied heavily on AI tools for academic purposes. This reliance was linked to an erosion of critical thinking and language skills, while senior students demonstrated more advanced and frequent use of AI tools compared to students in earlier academic years. In this context, a global survey across 16 countries revealed that 86% of students regularly use AI in their studies, with 54% using it weekly and over 66% employing it for information retrieval. Furthermore, 80% reported that AI in universities does not fully meet their expectations, 60% expressed concerns regarding data privacy, security, content reliability, and fairness of AI-based assessments, and 50% indicated that excessive reliance on AI would

negatively affect their academic performance and reduce the value of their education (Digital Education Council, 2024).

Recent international literature has increasingly examined the relationship between the use of artificial intelligence (AI) in higher education and students' critical thinking skills, reporting mixed and sometimes contradictory findings (Premkumar et al., 2024; Fischer et al., 2024; Kasneci et al., 2023; Cotton et al., 2023). While some studies suggest that AI-supported learning environments can enhance analytical reasoning and higher-order cognitive engagement by reducing cognitive load and supporting problem-solving processes, others warn that excessive or unregulated reliance on generative AI tools may weaken students' evaluative judgment, originality, and depth of reasoning (Premkumar et al., 2024; Fischer et al., 2024).

Within this debate, academic integrity emerges as a critical contextual factor that can shape how students interact with AI technologies, influencing whether AI use supports or undermines meaningful learning and critical thinking development (García-López and Trujillo-Liñán, 2025; Perkins, 2023). Academic integrity encompasses honesty, trust, fairness, responsibility, and ethical conduct, and determines whether students approach AI as a supportive learning tool or as a means to circumvent effort and academic rules (Mattar, 2022; Khatri and Karki, 2023). Students with high academic integrity are more likely to use AI responsibly, leveraging it to enhance analytical reasoning, understanding, and creativity. Conversely, students with low integrity may misuse AI, leading to plagiarism, reduced originality, and weakened critical thinking skills (Al Kaabi, 2025; Vieriu and Petrea, 2025; Uddin and Abu, 2024).

Despite its potential influence, limited empirical research has examined academic integrity as a moderating variable that conditions the impact of AI use on critical thinking, particularly among postgraduate students in non-Western higher education contexts (García-López and Trujillo-Liñán, 2025; Perkins, 2023). This gap justifies treating academic integrity as a moderator: it provides the conceptual rationale that the effectiveness of AI in fostering critical thinking is not uniform but depends on students' ethical engagement with AI tools. In other words, the relationship between AI usage and critical thinking may be stronger or more positive for students with high academic integrity, and weaker or even negative for students with low integrity.

Understanding how students engage with AI tools, and how academic integrity shapes this engagement, benefits from a theoretical perspective that considers attitudes, social expectations, and perceived behavioral control as key influences on academic behavior. In this regard, the Theory of Planned Behavior (TPB), developed by Ajzen (1991, 2002), offers a well-established conceptual lens for interpreting students' orientations toward the use of artificial intelligence (AI) in academic writing contexts.

According to TPB, individuals' engagement with a given behavior is informed by beliefs about its expected outcomes, which shape attitudes toward that behavior. Applied to AI-supported academic work, students' perceptions of the usefulness of AI tools—such as assisting with idea generation, enhancing writing coherence, or facilitating research processes—may influence their propensity to incorporate these tools into their academic practices. At the same time, normative beliefs derived from peers, instructors, and institutional expectations contribute to subjective norms that may either encourage or constrain AI use, particularly when ethical considerations and academic integrity are salient. Perceived behavioral control further reflects students' sense of capability in using AI effectively and responsibly, taking into account both individual competence and contextual constraints. Higher levels

of perceived control are likely to support more deliberate and reflective engagement with AI tools, including attention to originality and critical evaluation of AI-generated outputs. Conversely, limited perceived control may be associated with more superficial or uncritical reliance on AI, potentially affecting higher-order cognitive processes such as analysis, evaluation, and creativity. From this perspective, TPB highlights how the interaction of attitudes, subjective norms, and perceived behavioral control can inform patterns of AI engagement in academic settings, underscoring the importance of pedagogical approaches that promote ethical awareness and critical thinking (Ajzen, 1991, 2002).

Recent research in higher education contexts have indicated that attitudes toward AI, perceived social expectations, and perceived behavioral control remain influential factors in shaping intentions to use emerging educational technologies, including generative AI tools. Empirical studies drawing on TPB-related constructs have shown that positive attitudes toward AI, supportive academic norms, and higher perceived control are associated with greater willingness to adopt AI for academic purposes, as well as more responsible and ethical patterns of use (Ivanov et al., 2024; Ng and Ridzuan, 2025; Ngo et al., 2025). Collectively, these findings support the relevance of TPB as a theoretical reference point for interpreting relationships among AI usage, academic integrity, and critical thinking within contemporary educational environments.

Within the context of higher education in Jordan, institutions face increasing challenges due to the widespread use of artificial intelligence (AI) tools by students. Key issues include rising rates of academic plagiarism, a decline in the originality of submitted work, excessive reliance on these technologies, weakened critical thinking skills, and diminished quality of academic outputs. This situation has placed growing pressure on Jordanian universities to address these risks by developing strategies to curb unethical and irresponsible AI use, adopting advanced plagiarism detection software, enforcing disciplinary measures against students involved in academic dishonesty, and implementing awareness programs highlighting the impact of AI on students' cognitive skills (Gammoh, 2024).

At the faculty level, nearly half of academic staff (approximately 47%) report using AI tools for certain research and teaching tasks, while fewer than one-third possess sufficient knowledge of the ethical guidelines governing such use; around 70% expressed ethical concerns regarding plagiarism and exam misconduct (Alnsour et al., 2025). Among students, AI adoption is rapidly expanding, with roughly 90% reporting significant use of ChatGPT and a similar proportion employing AI tools for specific academic purposes. Although most students (about 87%) perceive these tools as educationally beneficial, the actual positive impact is more variable: approximately 58% reported improved academic performance, while around 40% indicated significant gains in comprehension (Al Mashagbeh et al., 2025). These findings underscore the necessity of implementing strict data privacy safeguards and adopting a comprehensive, well-considered approach to ensure ethical deployment of AI technologies within educational institutions (Ayasrah, 2025).

Critical thinking involves core cognitive processes, including analysis, evaluation, and reasoning, and is central to achieving logical conclusions and problem-solving. It relies on metacognitive skills such as attention to accuracy, relevance, and depth, and is essential for preparing students to be ethically responsible contributors to society (Dwyer et al., 2014; Bangun and Pragholaipati, 2021; Alwehaibi, 2012). Generative AI (GenAI) can support critical thinking by assisting with lower-order tasks like proofreading and initial idea generation, thereby reducing cognitive load and allowing focus on higher-order

analytical and creative work. However, excessive reliance may encourage superficial engagement and cognitive offloading, weakening deep cognitive involvement and critical evaluation (Premkumar et al., 2024; Fischer et al., 2024; Shukor and Osman, 2025; Essien et al., 2024).

Among postgraduate students, this dual effect is pronounced: AI may facilitate task completion but can also reduce evaluative reasoning and creativity, potentially fostering cognitive laziness and undermining adherence to ethical academic writing (Nguyen, 2025; Gonsalves, 2024). Irresponsible use threatens the quality of learning, diminishing language, analytical, and evaluative skills, and negatively impacting graduation readiness, research competence, and academic integrity (Al Kaabi, 2025; Vieriu and Petrea, 2025). Academic integrity is an ethical commitment encompassing honesty, trust, respect, fairness, responsibility, and courage, expected from both students and faculty (Mattar, 2022).

The proliferation of AI tools raises ethical concerns, including plagiarism, superficial understanding, bias, copyright issues, and diminished critical engagement, underscoring the need for responsible educational practices (Uddin and Abu, 2024). The threat to academic integrity is especially pronounced when AI-generated texts closely resemble human-authored work, facilitating plagiarism and misconduct, particularly if students fail to disclose AI usage (Khatiri and Karki, 2023).

Regulated AI use, combined with measures to curb overreliance, can enhance critical thinking and creativity while respecting research ethics and data privacy. Universities have responded by implementing plagiarism detection tools such as Grammarly and Turnitin, establishing clear policies, and providing specialized training, aiming to balance AI utilization with academic integrity (Al Kaabi, 2025; Alnsour et al., 2025; Khatiri and Karki, 2023). Despite the rapid adoption of generative AI in higher education, particularly for research and academic writing, knowledge about its long-term effects on higher-order thinking skills remains limited. Most studies focus on undergraduate students or general academic performance, without addressing advanced skills such as analysis, evaluation, and critical reasoning, especially among postgraduate students expected to demonstrate high levels of critical thinking and adherence to academic integrity (Kasneji et al., 2023; Cotton et al., 2023).

Accordingly, the primary gap addressed by this study is the absence of explanatory models integrating generative AI use with academic integrity as a moderating variable that shapes the relationship between AI utilization and the development of critical thinking skills. Current literature tends to treat academic integrity either as an independent ethical issue or as a potential risk associated with AI misuse, without examining its interactive role in influencing deep cognitive processes among postgraduate students (García-López and Trujillo-Liñán, 2025; Perkins, 2023). This study aims to investigate how the use of generative AI tools influences critical thinking skills among postgraduate students, and to examine the moderating role of academic integrity in this relationship. The study focuses on a context that has been underrepresented in the international literature, specifically postgraduate students in an Arab higher education setting.

Methods

Participant recruitment and data collection

A convenience sample of 555 postgraduate students from a public university located in Amman, the capital city, participated in this

study. The sample included both master's and doctoral students from humanities and scientific faculties. Female students comprised 52% of the sample, while male students comprised 48%. Regarding academic level, 63% of participants were master's students and 37% were PhD students. Participants' ages ranged from 25 to 47 years, with 41% aged 25 to 30, 37% aged 31 to 38, and 22% aged 39 to 47. All data collected reflect participants' self-reported perceptions and behaviors rather than objectively measured performance.

Data were collected during the 2025–2026 academic year using an online survey created with Google Forms. The survey was distributed to postgraduate students through social media, including WhatsApp groups for their courses, as well as their official university email accounts. Prior to beginning the survey, participants were presented with an informed consent statement describing the purpose of the study, the voluntary nature of participation, their right to withdraw at any time, and assurances of confidentiality, anonymity, and data protection. Continuing to complete the survey indicated their electronic consent. Consequently, all interpretations of AI usage, critical thinking, and academic integrity are considered students' self-perceptions.

The study employed a cross-sectional research design, allowing for the examination of relationships among artificial intelligence usage, critical thinking skills, and academic integrity within a single time frame without manipulating variables. Given the cross-sectional design, causal relationships among variables cannot be inferred. Ethical approval for this study was granted by the Ethics Committee of the Department of Sociology, University of Jordan (Ref: 32/2025). All procedures adhered to international standards for social science research, including the protection of participants' privacy and secure handling of the collected data.

Measures

Academic AI Usage Scale: Developed by [Chakraborty and Subramani \(2025\)](#), this scale consists of 24 items designed to assess postgraduate students' patterns and levels of artificial intelligence use in academic contexts. Confirmatory factor analysis supported a three-dimensional structure. In the present study, the original factor structure was retained, while minor terminological refinements were introduced to enhance conceptual clarity and contextual relevance for postgraduate research activities, without altering item content or measurement properties. The three dimensions are described as follows:

- 1 **Dependence (Research Support):** This dimension corresponds to the original "AI Dependence" factor and captures habitual reliance on AI for task completion, content generation, and verification. The label "Research Support" was adopted in the current study to better reflect its function within research-related academic tasks.
- 2 **AI for Academic Support (Academic Support):** This dimension aligns with the original "Academic Support" factor and captures the integration of AI tools into academic workflows for practical assistance, such as comprehension, organization, and time management. The terminology was retained for consistency with the study context.
- 3 **AI for Academic Skills (Learning and Study Aid):** This dimension corresponds to the original "AI for Academic Skills" factor and reflects constructive use of AI for developing transferable skills, including resource discovery and content visualization. The label "Learning & Study Aid" was used to emphasize its focus on learning processes and skill

development.

Critical Thinking Questionnaire (CTQ): Critical thinking skills were assessed using the CTQ developed by [Kobylarek et al. \(2022\)](#). The instrument includes 25 items measuring six core cognitive dimensions derived from Bloom's taxonomy: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating, in addition to an overall critical thinking score. Previous studies reported high internal consistency for the scale, with a Cronbach's alpha coefficient of 0.87. Items were rated on a five-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree).

AI Academic Integrity Scale for Postgraduate Students: Developed by [Maharmah et al. \(2025\)](#). The scale was initially developed with 24 items and subsequently refined through psychometric validation procedures to a final version of 17 items. It assesses three dimensions: Ethical Use of AI, Awareness of Misuse Risks, and Academic Writing Support. Responses were measured using a five-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree).

Statistical analysis

Data analysis was conducted using IBM SPSS Statistics (Version 25) for preliminary analyses and IBM AMOS (Version 26) for confirmatory factor analysis (CFA) and structural equation modeling (SEM). Descriptive statistics were calculated to examine the distributional properties of the study variables and to ensure that assumptions of normality were reasonably met prior to multivariate analysis. The internal consistency of all measurement instruments was evaluated using Cronbach's alpha coefficients to assess reliability. Construct validity was examined through Pearson correlation analyses among the study variables and their respective dimensions to provide evidence of convergent and discriminant validity. Given that all variables were assessed using self-report measures collected at a single time point, potential common-method bias was evaluated using Harman's single-factor test. This procedure examined whether a single latent factor accounted for the majority of variance across measurement items. In addition, confirmatory factor analysis (CFA) was conducted to verify the factorial structure of the Academic AI Usage Scale, the Critical Thinking Questionnaire, and the Academic Integrity Scale, and to ensure that the constructs were empirically distinct prior to testing structural relationships.

Structural equation modeling (SEM) was subsequently employed to examine the hypothesized relationships among AI usage, critical thinking, and academic integrity. Academic integrity was tested as a moderator using a latent interaction approach within SEM, with all variables modeled as continuous latent constructs represented by their respective dimensions. Because moderation was examined through a latent interaction model rather than formal multi-group comparisons, measurement invariance testing was not required. This analytic strategy allowed for the estimation of conditional effects while preserving the continuous nature of the moderator.

Results

Descriptive statistics and reliability

Descriptive statistics were computed for all study variables to examine their central tendency and distributional characteristics. The

results indicated acceptable levels of normality across all variables, with skewness values ranging from -0.45 to 0.32 and kurtosis values ranging from -0.38 to 0.41 . Reliability analyses demonstrated high internal consistency for all measurement instruments. The Academic AI Usage Scale showed strong reliability for the total scale ($\alpha = 0.91$), with Cronbach's alpha values of 0.88 for Dependence (Research Support), 0.87 for AI for Academic Support (Academic Support), and 0.85 for AI for Academic Skills (Learning & Study Aid).

The Critical Thinking Questionnaire also exhibited excellent reliability for the total scale ($\alpha = 0.92$), with acceptable reliability across its six dimensions (α values ranging from 0.84 to 0.89).

Similarly, the Academic Integrity Scale demonstrated high internal consistency for the total scale ($\alpha = 0.90$), with alpha coefficients of 0.88 for Ethical Use of AI, 0.86 for Awareness of Misuse Risks, and 0.85 for Support for Academic Writing. These results indicate that all scales possess satisfactory reliability for subsequent analyses.

Correlation analysis

Pearson correlation coefficients were computed to examine the relationships among AI usage dimensions, critical thinking dimensions, and academic integrity dimensions. As shown in Table 1, AI usage dimensions were positively and significantly associated with critical thinking dimensions, particularly lower-order cognitive skills (remembering, understanding, and applying), with correlation coefficients ranging from $r = 0.18$ to $r = 0.24$ ($p < 0.01$). Associations between AI usage and higher-order critical thinking skills (analyzing, evaluating, and creating) were weaker but remained statistically significant in most cases (r values ranging from 0.13 to 0.17).

In contrast, AI usage dimensions were negatively associated with academic integrity dimensions, with significant correlations observed across ethical use of AI, awareness of misuse risks, and support for academic writing (r values ranging from -0.12 to -0.22 , $p < 0.01$). Overall, the correlation results suggest that greater perceived use of AI tools is associated with higher perceived critical thinking skills, particularly at lower cognitive levels, while also being inversely related to perceptions of academic integrity.

Measurement model: confirmatory factor analysis

To examine the factorial validity of the study instruments and to ensure empirical distinctiveness among constructs, Confirmatory Factor Analysis (CFA) was conducted for each scale separately. The Academic AI Usage Scale, specified as a three-factor model (Dependence [Research Support], AI for Academic Support [Academic Support], and AI for Academic Skills [Learning and Study Aid]), demonstrated good model fit (CFI = 0.96 , TLI = 0.95 , RMSEA = 0.042 , SRMR = 0.035). The Critical Thinking Questionnaire, modeled with six factors (Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating), also showed acceptable fit to the data (CFI = 0.95 , TLI = 0.94 , RMSEA = 0.043 , SRMR = 0.036). Similarly, the Academic Integrity Scale, specified as a three-factor model (Ethical Use of AI, Awareness of Misuse Risks, and Support for Academic Writing), demonstrated good fit (CFI = 0.95 , TLI = 0.94 , RMSEA = 0.045 , SRMR = 0.038). These CFA results confirm that the predictor (AI usage), outcome (critical thinking), and moderator (academic integrity) constructs are empirically distinct, supporting their inclusion in the structural model.

TABLE 1 Correlations between AI usage dimensions, critical thinking dimensions, and academic integrity dimensions.

Predictor (AI usage)	Outcome (Critical thinking)	<i>R</i>	Moderator (Academic integrity)	<i>r</i>
Dependence (Research support)	Remembering	0.24**	Ethical Use of AI	-0.15**
	Understanding	0.22**	Awareness of Misuse Risks	-0.20**
	Applying	0.20**	Support for Academic Writing	-0.22**
	Analyzing	0.16*	—	—
	Evaluating	0.15*	—	—
	Creating	0.14*	—	—
Academic support	Remembering	0.21**	Ethical Use of AI	-0.12*
	Understanding	0.19**	Awareness of Misuse Risks	-0.18**
	Applying	0.18**	Support for Academic Writing	-0.20**
	Analyzing	0.13*	—	—
	Evaluating	0.12*	—	—
	Creating	0.11	—	—
Academic skills (Learning and Study aid)	Remembering	0.22**	Ethical Use of AI	-0.14*
	Understanding	0.21**	Awareness of Misuse Risks	-0.19**
	Applying	0.19**	Support for Academic Writing	-0.21**
	Analyzing	0.17*	—	—
	Evaluating	0.15*	—	—
	Creating	0.14*	—	—

* $p < 0.01$, ** $p < 0.001$. Correlations are among latent dimensions of study constructs. Predictor = AI Usage, Outcome = Critical Thinking, Moderator = Academic Integrity.

Structural model and moderation analysis

Structural Equation Modeling (SEM) was conducted to examine the hypothesized relationships between AI usage and critical thinking, as well as the moderating role of academic integrity. All constructs were modeled as latent variables represented by their respective dimensions. The overall structural model demonstrated acceptable fit to the data (CFI = 0.94 , TLI = 0.92 , RMSEA = 0.045 , SRMR = 0.038). Results indicated that AI usage was positively associated with lower-order critical thinking skills, including remembering ($\beta = 0.24$, $p < 0.001$), understanding ($\beta = 0.22$, $p < 0.001$), and applying ($\beta = 0.20$, $p < 0.001$). Associations with higher-order skills were comparatively weaker but remained statistically significant (analyzing: $\beta = 0.16$, $p < 0.01$; evaluating: $\beta = 0.15$, $p < 0.01$; creating: $\beta = 0.14$, $p < 0.01$).

To test the moderating role of academic integrity, a latent interaction approach was employed. The interaction term between AI usage

and academic integrity was statistically significant ($\beta = 0.18$, $p < 0.001$), indicating that academic integrity moderates the relationship between AI usage and critical thinking. Conditional effects (simple slopes) analysis revealed that the positive association between AI usage and critical thinking skills was stronger at higher levels of academic integrity (+1 SD), whereas weaker or non-significant associations were observed at lower levels of academic integrity (−1 SD), particularly for higher-order cognitive skills. These conditional estimates are derived from the continuous latent interaction model and are presented for illustrative purposes only; no formal group comparisons were conducted. Detailed conditional estimates are presented in Table 2.

To ensure factorial validity and discriminant properties for all three scales, Confirmatory Factor Analysis (CFA) was conducted. The Academic AI Usage Scale [3 factors: Dependence (Research Support), AI for Academic Support (Academic Support), AI for Academic Skills (Learning and Study Aid) showed good fit: CFI = 0.96, TLI = 0.95, RMSEA = 0.042, SRMR = 0.035]. The Critical Thinking Questionnaire (6 factors: Remembering, Understanding, Applying, Analyzing, Evaluating, Creating) demonstrated acceptable fit: CFI = 0.95, TLI = 0.94, RMSEA = 0.043, SRMR = 0.036. The AI Academic Integrity Scale (3 factors: Ethical Use of AI, Awareness of Misuse Risks, Support for Academic Writing) also showed good fit: CFI = 0.95, TLI = 0.94, RMSEA = 0.045, SRMR = 0.038. These results confirm that the predictor (AI Usage) and moderator (Academic Integrity) constructs are empirically distinct, reducing concerns of construct contamination.

Structural Equation Modeling (SEM) using IBM AMOS (Version 26) was conducted to examine whether academic integrity moderates the relationship between AI usage and critical thinking skills. All

constructs were modeled as latent variables represented by their respective dimensions, and a latent interaction approach tested the moderation effect. Model fit indices indicated acceptable fit: CFI = 0.94, TLI = 0.92, RMSEA = 0.045, SRMR = 0.038.

SEM results revealed that AI usage positively predicted lower-order critical thinking skills (Remembering: $\beta = 0.24$, $p < 0.001$; Understanding: $\beta = 0.22$, $p < 0.001$; Applying: $\beta = 0.20$, $p < 0.001$), whereas associations with higher-order skills were weaker (Analyzing: $\beta = 0.16$, $p < 0.01$; Evaluating: $\beta = 0.15$, $p < 0.01$; Creating: $\beta = 0.14$, $p < 0.01$). The latent interaction analysis indicated that academic integrity significantly moderated these relationships (interaction $\beta = 0.18$, $p < 0.001$). Conditional effects (simple slopes) indicated that the positive association between AI usage and critical thinking was stronger at higher levels of academic integrity (+1 SD), whereas weaker or nonsignificant associations emerged at lower levels (−1 SD), particularly for higher-order skills (see Table 2 and Figure 1).

Discussion

The study findings reveal nuanced relationships between AI usage and critical thinking skills among postgraduate students. Lower-order cognitive skills—remembering, understanding, and applying—benefit consistently from AI usage, whereas higher-order skills such as analyzing, evaluating, and creating show weaker direct associations. This pattern suggests that while AI tools facilitate initial stages of knowledge acquisition, their impact on more complex cognitive processes may be limited unless accompanied by ethical and deliberate engagement. Importantly, AI usage dimensions were negatively associated

TABLE 2 Conditional effects (Simple slopes) of AI usage on critical thinking at low (−1 SD) and High (+1 SD) levels of academic integrity.

AI usage dimension	Critical thinking dimension	β (Low Integrity −1 SD)	p	β (High Integrity +1 SD)	p
Dependence (Research support)	Remembering	0.12	0.004	0.28	<0.001
	Understanding	0.10	0.014	0.25	<0.001
	Applying	0.09	0.027	0.23	<0.001
	Analyzing	−0.03	0.434	0.15	0.002
	Evaluating	−0.04	0.343	0.14	0.004
	Creating	−0.05	0.223	0.12	0.014
Academic support	Remembering	0.11	0.008	0.26	<0.001
	Understanding	0.09	0.020	0.24	<0.001
	Applying	0.08	0.043	0.22	<0.001
	Analyzing	−0.02	0.556	0.13	0.007
	Evaluating	−0.03	0.437	0.12	0.014
	Creating	−0.04	0.293	0.11	0.019
Academic skills (Learning and Study aid)	Remembering	0.10	0.011	0.27	<0.001
	Understanding	0.09	0.020	0.25	<0.001
	Applying	0.08	0.042	0.23	<0.001
	Analyzing	−0.01	0.742	0.14	0.004
	Evaluating	−0.02	0.503	0.13	0.007
	Creating	−0.03	0.378	0.12	0.014

Low and high levels reflect conditional values (simple slopes) at −1 SD and +1 SD of academic integrity. No group-based comparisons were conducted.

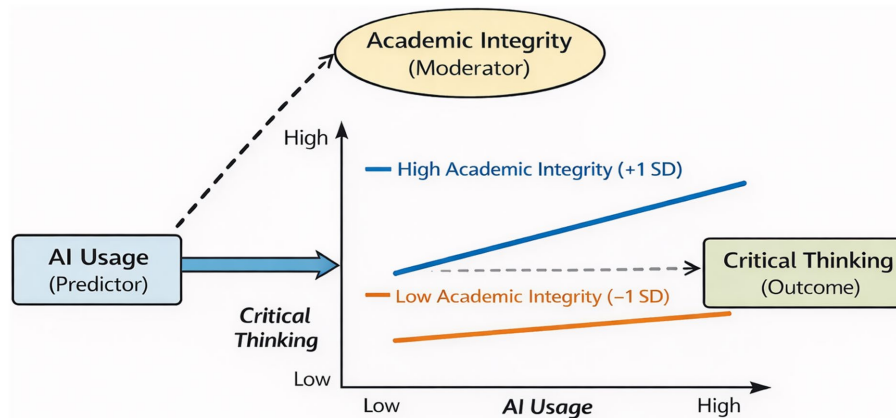


FIGURE 1

Conceptual illustration of the moderating role of academic integrity. Conditional level (+1 SD and -1 SD) of academic integrity are shown for illustrative purposes; no group- based comparisons were included.

with academic integrity dimensions, particularly with awareness of misuse risks and support for academic writing, highlighting the potential for responsible AI engagement to amplify cognitive gains. These observations underscore the moderating role of academic integrity: students adhering to ethical standards can leverage AI tools to achieve broader cognitive benefits, whereas those with lower integrity may experience limited or uneven advantages. The interpretation of these patterns aligns with Bloom's Taxonomy, situating AI's contribution within a hierarchical framework of cognitive development. AI facilitates lower-order cognitive processes, enabling students to allocate more cognitive resources toward higher-order analytical and creative tasks, provided that ethical considerations and responsible use guide their engagement. In this sense, academic integrity functions as a crucial contextual factor shaping the depth and quality of critical thinking outcomes. Thus, rather than repeating the raw correlation values presented in the Results, the discussion emphasizes interpretation, conceptual framing, and the role of academic integrity as a moderator, setting the stage for subsequent sections that analyze detailed paths, latent interactions, and implications for practice and policy.

Among postgraduate students, the use of AI tools is recurrent in tasks such as idea development, spell-checking, paraphrasing, improving coherence, as well as tools that assist in organizing related academic papers. Despite their significance, student's express concerns about the potential negative effects on critical thinking and creativity, and their intention to use these tools is strongly associated with personal standards and influenced by the surrounding environment (Anani et al., 2025). The AI revolution and its technologies have transformed higher education, placing a significant burden on institutions to address challenges and adopt all possible strategies to enhance their performance in order to remain competitive. Amid this revolution, higher education institutions face challenges in effectively fostering students' critical thinking skills. Despite widespread recognition of its importance for success in the labor market, a gap persists between the acknowledged value of critical thinking and the actual proficiency levels of university students (Hamail, 2023; Premkumar et al., 2024; Zoghbar, 2025).

Several international and regional studies provide additional context for these findings. Al Kaabi (2025) revealed that more than 75% of students used AI programs for academic purposes, and 94.5% of them exhibited deterioration in critical thinking and language skills

due to this reliance. Similarly, Aldossary et al. (2024) reported that students hold positive acceptance and high perceptions regarding the expected value of generative AI in education. Despite recognizing the limitations of these tools, students showed a lack of awareness regarding the importance of ethical compliance when using them. The study also indicated that generative AI has a positive impact on achieving sustainable development goals in education. In Saudi Arabia, Almulla and Ibrahim Ali (2024) found that postgraduate students used ChatGPT for various academic purposes, including assistance with homework and research. About 60% of students reported positive effects on their academic performance and research skills. The study also revealed a significant relationship between the frequency of ChatGPT use, students' satisfaction and perceived usefulness, and its effectiveness in enhancing learning experiences, while also noting a negative impact on critical thinking skills. In the Jordanian academic context, Ayasrah (2025) identified a strong correlation between ethical concerns regarding AI integration and data privacy precautions. The study emphasized the need to establish comprehensive ethical standards to address issues such as algorithmic transparency, bias reduction, and accountability measures. It also highlighted the importance of AI systems in education being transparent and interpretable, and of evaluating these systems in ways that enhance inclusivity and protect data privacy variables, including encryption and cybersecurity.

To contextualize the observed effects of AI usage on cognitive skills, Bloom's Taxonomy, developed by Benjamin Bloom and colleagues (Bloom et al., 1956), classifies cognitive skills hierarchically, ranging from lower-order skills such as remembering and understanding to higher-order skills including analyzing, evaluating, and creating. This framework is widely used in educational research to design assessments and evaluate the depth of learning outcomes. In line with this framework, the path analysis in the current study revealed that AI usage directly enhances lower-order cognitive skills—remembering, understanding, and applying—while its direct effects on higher-order skills, such as analyzing, evaluating, and creating, are comparatively weaker.

Importantly, the latent interaction analysis demonstrated that academic integrity serves as a significant moderating factor. Students with higher academic integrity were able to leverage AI usage more effectively across all dimensions achieving stronger gains in higher-order critical thinking skills. In contrast, students with lower integrity

primarily benefited in lower-order skills, with minimal improvement in higher-order skills. These findings indicate that responsible and ethical use of AI tools can support students' progression across Bloom's hierarchical levels, from basic knowledge acquisition to complex analytical and creative thinking. This aligns with previous research showing that structured and guided AI integration enhances both foundational and advanced cognitive skills (Essien et al., 2024).

In a study conducted by Gonsalves (2024) aimed at examining how the integration of AI tools into marketing education assessments affects the critical thinking skills of master's students in marketing, the results showed that students interacted with AI across Bloom's cognitive, affective, and metacognitive domains, as well as analytical skills. Bloom's Taxonomy was proposed as a framework for AI-enhanced learning, encompassing nonlinear learning, strategic thinking, and ethical reasoning. The Kosovo study provides valuable insights into students' AI use in higher education. It showed that greater engagement with AI tools is associated with more positive attitudes and responsible usage, while lack of training or familiarity with institutional policies creates challenges. These findings support the idea that academic integrity can moderate the relationship between critical thinking and AI use, as adherence to ethical standards and guidelines enhances responsible adoption. The study also indicated that demographic factors had little influence, highlighting the stronger role of experience, social norms, and institutional support (Çerkini et al., 2025).

Moreover, the present findings provide a nuanced understanding of how generative AI usage relates to critical thinking skills among postgraduate students. Path analysis revealed that AI usage directly enhances lower-order cognitive skills—remembering, understanding, and applying—while its direct effects on higher-order skills such as analyzing, evaluating, and creating are comparatively weaker. Importantly, latent interaction analysis demonstrated that academic integrity serves as a significant moderating factor: the positive effects of AI usage on all critical thinking dimensions were substantially stronger for students with higher integrity compared to those with lower integrity. For instance, Academic Support had a pronounced impact on higher-order skills among high-integrity students, whereas its effect was negligible or slightly negative for low-integrity students. Similar patterns emerged for Academic Support and Academic Skills (Learning and Study Aid), highlighting that ethical engagement with AI amplifies cognitive benefits while mitigating potential negative consequences of excessive reliance. Conceptually, academic integrity can be understood as a moderator that shapes the extent to which AI usage translates into meaningful cognitive gains. Students with high integrity are more likely to engage deeply with AI outputs, critically evaluating and integrating information, whereas those with lower integrity may rely excessively on AI, resulting in superficial engagement and reduced critical thinking performance (Uddin and Abu, 2024; Khatri and Karki, 2023; Kasneci et al., 2023). According to Kalniņa et al. (2024) the Low adoption of AI was observed among pre-service teachers, despite recognition of its potential benefits, including language support and access to global knowledge. Challenges such as reduced critical thinking and concerns over plagiarism were also highlighted. These results align with the notion of academic integrity as a moderating factor: students' adherence to ethical standards and institutional guidelines may strengthen the positive effects of critical thinking on responsible AI use. Furthermore, demographic variables had little influence on AI adoption, emphasizing that experience, social norms, and institutional support are more decisive.

Thus, the moderating role of academic integrity indicates that the benefits or risks of AI usage depend on students' adherence to ethical standards, emphasizing the need for interventions—such as ethical training, monitoring, and institutional guidelines—to ensure responsible AI integration. This perspective provides a theoretical and empirical basis for understanding differential effects of AI on critical thinking in postgraduate education, addressing gaps in current research that largely focus on undergraduate populations or general academic performance (Cotton et al., 2023).

Finally, our findings have several important implications for future research, educational practice, and policy in higher education. The results suggest that integrating AI tools into postgraduate learning can enhance both lower-order and higher-order cognitive skills when accompanied by ethical guidance and strong academic integrity. Educational institutions should consider designing structured interventions, including ethical training, monitoring, and clear guidelines for responsible AI use, to maximize the cognitive benefits and minimize potential risks such as superficial engagement or reduced critical thinking. These findings also point to the potential for AI-enhanced learning frameworks to be applied across diverse disciplines, promoting strategic, metacognitive, and creative thinking among students.

However, several limitations should be acknowledged. First, the study employed a cross-sectional design, which prevents causal inferences and only allows for associations to be observed. Second, the sample was selected from a single public university and included both master's and doctoral students. While this approach ensured sufficient statistical power for structural equation modeling, it may limit the generalizability of the findings to other higher education contexts, and future research could examine these relationships separately for master's and doctoral students to provide more nuanced insights. Third, reliance on self-reported measures may introduce biases such as social desirability or inaccurate reporting of AI usage and critical thinking skills. Future research could also address these limitations by employing longitudinal or experimental designs, expanding samples to multiple universities, and incorporating objective measures of AI engagement and cognitive performance. Despite these limitations, the present study provides a comprehensive understanding of how AI usage and academic integrity interact to influence critical thinking, offering a foundation for both theoretical development and practical applications in higher education.

Conclusion

This study examined how AI usage influences critical thinking skills among postgraduate students, highlighting the moderating role of academic integrity. Results show that AI tools effectively enhance lower-order cognitive skills (remembering, understanding, applying), while higher-order skills (analyzing, evaluating, creating) benefit more selectively. Academic integrity plays a crucial moderating role: students with higher integrity leverage AI to achieve stronger gains across all critical thinking dimensions, whereas students with lower integrity show limited improvement, mainly in lower-order skills. These findings underscore the importance of ethical guidance and institutional measures—such as structured training, monitoring, and clear usage guidelines—to maximize the benefits of AI integration while mitigating potential risks of overreliance. Despite limitations related to the cross-sectional design, self-reported measures, and a single-university

sample, the study provides practical and theoretical insights into the interaction between AI usage and academic integrity in shaping cognitive outcomes. Future research should explore longitudinal designs, diverse populations, and objective measures of AI engagement and learning performance.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author/s.

Ethics statement

Ethical approval for this study was granted by the Scientific Committee of the Department of Sociology at the University of Jordan, Ref: 32/2025. The study was conducted in accordance with local legislation and institutional requirements. Written informed consent was not required from the participants in accordance with national legislation and institutional requirements.

Author contributions

RALK: Conceptualization, Methodology, Resources, Writing – original draft, Writing – review & editing. RALs: Conceptualization, Formal analysis, Methodology, Resources, Supervision, Writing – original draft, Writing – review & editing. RALr: Writing – review & editing.

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